

Causal Inference Workshop

Week 5 - Difference-in-Differences, Triple Differences, TWFE, ...

Causal Inference Workshop

February 18, 2024

Workshop outline

A. Causal inference fundamentals

- Modeling assumptions matter too
- Conceptual framework (potential outcomes framework)

B. Design stage: common identification strategies

- IV + RDD [coding]
- Event Studies, DiD, DiDiD, New TWFE Lit [coding]
- Synthetic Control / Synthetic DiD [coding]

C. Analysis stage: strengthening inferences

- Limitations of identification strategies, pre-estimation steps
- Estimation [controls] and post-estimation steps [supporting assumptions]

D. Other topics in causal inference and sustainable development

- Inference (randomization inference, bootstrapping)
- Weather data regressions, other common/fun SDev topics [coding]
- Remote sensing data, other common/fun SDev topics

Causal inference roadmap

- *Potential outcomes* [framework]
 - Causal effect is the difference between two potential outcomes
 - We can't observe this difference, but can see differences in average observed outcomes
 - If **(conditional) independence assumption** holds, can estimate unbiased ATT
- *Identification* [application/implementation] [last week, and today, ... and next week!]
 - In most empirical settings, IA and CIA do not hold, which is why we need an **identification strategy**
 - Want to eliminate selection bias (identification problem)
- *Estimation* [application/implementation]
 - (Usually) use linear regression model
 - $\hat{\beta}_{OLS}$ unbiased estimator for ATT if e is uncorrelated with treatment (regression problem)

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Overview

Event studies

Difference-in-differences (DiD)

Triple differences (DiDiD)

Two-Way Fixed Effects (TWFE)

Overview of event studies, DiD+, and TWFEs

- General setup:
 - **Objective:** estimate the impact of some treatment at a certain time
 - **Data:** panel data (multiple units, multiple time periods)
- to estimate **causal** effect of event, we want the **counterfactual** of it occurring (this is the unit's outcome had the event not occurred)

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 - **DiD, DiDiD, TWFE:** Some units never get treated, assume CF *trends* of treated and untreated are parallel, use control units to remove trends in *Y* in treated (*within* and *between* variation)
 - **Synthetic control, synthetic DiD:** Units large aggregates, most (some) units never get treated, combine untreated units to construct optimal synthetic unit that captures CF trajectory of treated unit(s)

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Event study, DGP and identifying assumptions

- Want to estimate the causal effect of an *event*, which occurs at time τ and affects *all units* in the population, on some outcome Y
(treatment assignment is a function of the period)
 - Identifying assumptions
- | | |
|--------------------------------|--|
| A1. exogeneity (random timing) | event unpredictable, not result of Y |
| A2. sample composition | does not vary over time |

Event study, estimand and estimator

- Estimand

$$\begin{aligned}\beta_{ES}^{\ell} &= \mathbb{E}[Y_{it}|t = \tau + \ell] - \mathbb{E}[Y_{it}|t = \tau - 1] \\ &= \mathbb{E}[Y_{i,\tau+\ell}^1] - \mathbb{E}[Y_{i,\tau-1}^0] \\ &= \underbrace{\mathbb{E}[Y_{i,\tau+\ell}^1] - \mathbb{E}[Y_{i,\tau+\ell}^0]}_{ATT^{\ell}}\end{aligned}$$

Key assumption used: $\mathbb{E}[Y_{i,\tau+\ell}^0] = \mathbb{E}[Y_{i,\tau-1}^0]$ because of exogeneity of τ .

- Estimator

- $\hat{\beta}_{OLS}$ of the regression on a set of time dummies (omitting the period before the event to normalize it as 0) consistently estimates β_{ES}

$$Y_{it} = \sum_{t=-K}^{\tau-2} [\beta_t \mathbb{1}\{t\}] + \beta \mathbb{1}\{\tau\} + \sum_{t=\tau+1}^L [\beta_t \mathbb{1}\{t\}] + e_{it}$$

Event study, some economics examples

- Many finance studies with very high frequency stock market data
- Economic effects of hospitalization (Dobkin et al. 2018, AER)
 - Look at the effects of hospitalization on various metrics (unpaid bills, earnings, credit)
 - Non-parametric event study

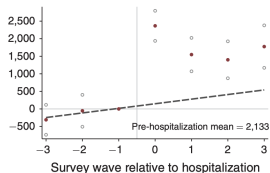
$$y_{it} = \gamma_t + \alpha X_{it} + \sum_{r=-3}^{-2} \mu_r \mathbb{1}\{r\} + \sum_{r=0}^3 \mu_r \mathbb{1}\{r\} + \epsilon_{it}$$

- *Identifying assumption*: conditional on having a hospital admission during the observation window and the included controls, the timing of the admission is uncorrelated with the outcome.
→ an admission that is preceded by deteriorating health, or an admission caused by the adverse health effects of job loss would violate this assumption.

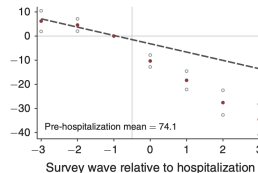
Event study, some economics examples

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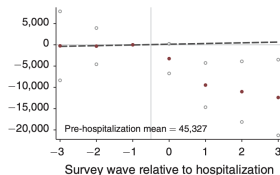
Panel A. Out-of-pocket medical spending



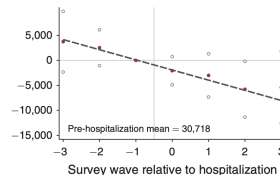
Panel B. Working part- or full-time



Panel C. Respondent earnings



Panel D. Spousal earnings



Event study, some economics examples

- Many finance studies with very high frequency stock market data
- Economic effects of hospitalization (Dobkin et al. 2018, AER)
 - Look at nonparametric event study, then main parametric specification has **linear trend** in event time (based on nonparametric results)

$$y_{it} = \gamma_t + \alpha X_{it} + \delta r + \sum_{r=0}^3 \mu_r \mathbb{1}\{r\} + \epsilon_{it}$$

- *Identifying assumption*: conditional on having a hospital admission and the included controls, the timing of the admission is uncorrelated with deviation of the outcome from a linear trend in event time. $\mathbb{E}[Y_{i,\tau+\ell}^0] - \delta\ell = \mathbb{E}[Y_{i,\tau-1}^0] - \delta$
→ This assumption requires that there are no factors correlated with y_{it} that, conditional on the included controls, occur contemporaneously with the hospital admission – such as a job loss that simultaneously produces a hospital admission.

Event study, best practices, strengths, and weaknesses

- Best practices
 - Plot and report all β_t 's, to check that they are not changing up to the event (change would suggest presence of pre-trends, which makes it hard to interpret the event in the absence of a sharp trend discontinuity)
 - As in any observational study, adjust for all other relevant pre-treatment variables
- Strengths & weaknesses
 - + Flexible, allows looking at whether effects are dynamic
 - Often empirical papers look at: 1) effect on average, 2) pre-trends, and 3) dynamics
 - May be difficult to rule out other things changing at the same time (unobserved confounders... the rooster concluding the sun rises because of his crowing?)

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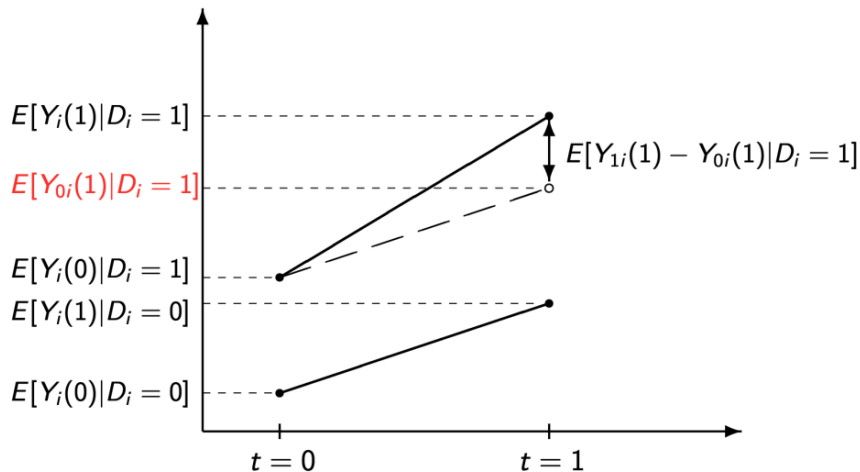
DiD, DGP and identifying assumptions

- Treatment assignment/exposure is now a function of two dimensions: group (treatment/control) and most commonly time (pre/post) exposure
 - $G_i = \mathbb{1}\{i \in \text{treatment group}\}$ and $P_t = \mathbb{1}\{t \in \text{post period}\} = \mathbb{1}\{t \geq \tau\}$
→ compare *change over time* in Y in treated to *change over time* in Y in control
- Identifying assumptions

A1. “parallel” counterfactual trends	$\mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_i = 1] = \mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_i = 0]$
A2. group composition	does not vary over time

DiD, DGP and identifying assumptions

$Y_i(t)$: outcome of i at time t . $Y_{di}(t)$: potential outcome under d of i at time t .



DiD, some classic examples

- 1850s cholera death rates (Snow 1855, table from [The Effect Book](#))

Region Supplier	Death Rates 1849	Death Rates 1854
Non-Lambeth Only (Dirty)	134.9	146.6
Lambeth + Others (Mix Dirty and Clean)	130.1	84.9
Death rates are deaths per 10,000 1851 population.		

- Fast food industry minimum wages (Card and Krueger [1994](#), AER)

DiD, some classic examples

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DiD, some classic examples

- 1850s cholera death rates (Snow 1855, table from [The Effect Book](#))
- Fast food industry minimum wages (Card and Krueger 1994, AER)
 - Examine impact of minimum wage on employment (NJ vs. PA, fast food)

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE
IN NEW JERSEY MINIMUM WAGE

Variable	Stores by state			Stores in New Jersey ^a			Differences within NJ ^b	
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low– high (vii)	Midrange– high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	–2.69 (1.37)	–2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	–2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	–2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores ^c	–2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	–2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0 ^d	–2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	–2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

DiD, estimand and estimator

- Estimand

$$\begin{aligned}\beta_{DID} &= \left(\mathbb{E}[Y_{i,1} | G_i = 1] - \mathbb{E}[Y_{i,0} | G_i = 1] \right) - \left(\mathbb{E}[Y_{i,1} | G_i = 0] - \mathbb{E}[Y_{i,0} | G_i = 0] \right) \\ &= \dots \\ &= \underbrace{\mathbb{E}[Y_{i,1}^1 - Y_{i,1}^0 | G_i = 1]}_{ATT \text{ (in post period)}}\end{aligned}$$

- Estimator

$$Y_{it} = \alpha + \beta_G G_i + \beta_P P_t + \beta_{DID} G_i P_t + e_{it}$$

With two groups and two periods, $\beta_{DID} = \beta_{FE}$ for the below (to be continued...!)

$$Y_{it} = \lambda_G + \delta_P + \beta_{FE} G_i P_T + e_{it}$$

DiD, best practices, strengths, and weaknesses

- Best practices
 - Support assumption of parallel counterfactual trends by showing that pre-treatment trends coincide (and estimate “dynamic” version of model and check that coefficients for time periods before treatment are zero)
 - Adjust for other relevant pre-treatment confounders
- Strengths & weaknesses
 - + Repeated observations get rid of unobserved time-invariant confounders
 - + Unobserved trends not a problem (unlike event studies) as long as trends of the groups are parallel (so that you can use the trends of the untreated outcome of the control group to fill in the missing unobserved trend of the treated group)
 - + Repeated cross-sectional ok, as long as sample composition doesn't vary over time
 - Anything beyond simple binary DiD has potential issues that need to be addressed, as we'll soon discuss

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DiDiD, DGP and identifying assumptions

- Treatment also varies along a third dimension or “subgroup” (e.g., gender or space)
 - $S_i = \mathbb{1}\{i \in \text{treatment in dimension 3}\}$
- Identifying assumptions

A1. “parallel” counterfactual trends across subgroups	$\mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_1, S_1] - \mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_0, S_1] =$ $\mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_1, S_0] - \mathbb{E}[Y_{i,1}^0 - Y_{i,0} G_0, S_0]$
A2. subgroup composition	does not vary over time

DiDiD, estimand and estimator

- Estimand

$$\begin{aligned}\beta_{DIDID} &= \left[(\mathbb{E}[Y_{i,1}|G_1, S_1] - \mathbb{E}[Y_{i,0}|G_1, S_1]) - (\mathbb{E}[Y_{i,1}|G_0, S_1] - \mathbb{E}[Y_{i,0}|G_0, S_1]) \right] - \\ &\quad \left[(\mathbb{E}[Y_{i,1}|G_1, S_0] - \mathbb{E}[Y_{i,0}|G_1, S_0]) + (\mathbb{E}[Y_{i,1}|G_0, S_0] - \mathbb{E}[Y_{i,0}|G_0, S_0]) \right] \\ &= \dots \\ &= \underbrace{\mathbb{E}[Y_{i,1}^1 - Y_{i,1}^0 | G_1, S_1]}_{ATT \text{ (in post period)}}\end{aligned}$$

- Estimator

$$\begin{aligned}Y_{it} &= \alpha + \beta_G G_i + \beta_S S_i + \beta_P P_t \\ &\quad \beta_{GS} G_i S_i + \beta_{GP} G_i P_t + \beta_{SP} S_i P_t + \\ &\quad \beta_{DIDID} G_i S_i P_t + e_{it}\end{aligned}$$

DiDiD, best practices, strengths, and weaknesses

- Best practices
 - Triple difference makes for a very specific control group; so justify why a double difference isn't satisfactory
 - As in any observational study, adjust for all other relevant pre-treatment variables
- Strengths & weaknesses
 - + Triple difference can difference out more confounding elements, so harder to find confounders
 - Requires more data and variation

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So, what's the deal with TWFE?...



TWFE problem overview

- Canonical 2x2 DiD (2 periods and 2 groups, treated groups experience treatment in period 2) **great** – easy to understand what is estimated and assumptions needed
- However, most actual research these days is **not** using 2x2 DiD, but rather TWFE:

$$y_{it} = \lambda_i + \delta_t + \beta_{TWFE} D_{it} + e_{it}$$

- Problems begin because we do not (or did not) actually understand how this estimator is comparing mean outcomes across groups
- **Big picture takeaway:** TWFE regressions estimate *weighted sums* of the ATE in each group and period, with weights that sum to one but may make TWFE estimator biased (and *may* even be negative)

TWFE problem overview

- **Big picture takeaway:** TWFE regressions estimate *weighted sums* of the ATE in each group and period, with weights that sum to one but may make TWFE estimator biased (and *may* even be negative)

$$y_{it} = \lambda_{g[i]} + \delta_t + \beta_{TWFE} D_{g[i]t} + e_{g[i]t}$$

→ $\hat{\beta}_{TWFE}$ is a specific weighted sum of the ATE in each treated (g,t) cell (Chaisemartin and D'Haultfœuille 2020)

$$\beta_{ATT} = \mathbb{E} \left[\sum_{(gt): D_{gt}=1} \frac{N_{gt}}{N_1} ATE_{gt} \right], \quad \mathbb{E}[\hat{\beta}_{TWFE}] = \mathbb{E} \left[\sum_{(gt): D_{gt}=1} \frac{N_{gt}}{N_1} w_{gt} ATE_{gt} \right]$$

→ weights may vary and can even be **negative!** (Goodman-Bacon 2021)
negative weights huge potential issue, as it may be $\hat{\beta}_{TWFE} < 0$ even if all $ATE_{gt} > 0$ (e.g., $1.5 \times 1 - 0.5 \times 4$...)

TWFE problem overview

- When do we need to be worried about these f***ing weights?
- How do we start digesting all the TWFE literature?
- What should we do if we want to use TWFE?
- Should we even use TWFE????
- ??????????????????????????
- AHHHHHHH!

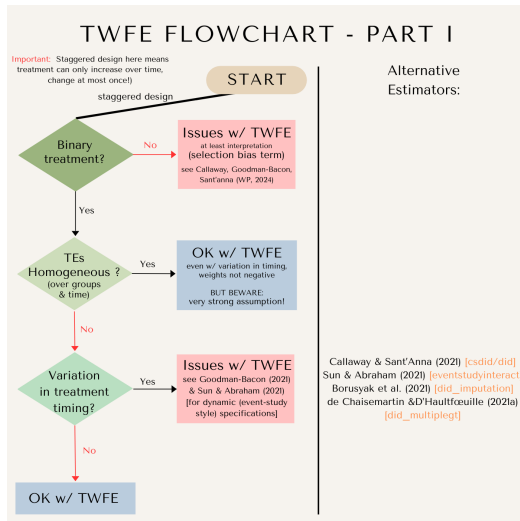
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- ↑ Pretty much the TWFE literature over the past x years...?

TWFE and when it's a problem

1. You should worry when **treatment effects are heterogeneous** (not constant across groups or over time) – this is *most* studies!
 2. You should also understand stronger assumptions with **continuous** treatments
 3. Definitional point: often people refer to “staggered” designs (technically means treatment can only increase, BUT people usually refer to binary treatments that can only increase, but can increase at any time as “staggered”)
- Suggested reading order:
 - Chaisemartin and D’Haultfœuille 2023 review: *Two-Way Fixed Effects and Differences-in-Differences with Heterogeneous Treatment Effects: A Survey*
 - Chaisemartin and D’Haultfœuille 2020 (AER, overview of problem & alternate estimator)
 - Goodman-Bacon (2021) helpful for understanding intuition behind negative weights and “forbidden comparisons” (using early-treated group as control for late-treated group)
→ then decide what else you need to learn about for your specific set-up!

TWFE and when it's a problem



TWFE and when it's a problem

Chaisemartin and D'Haultfœuille 2023

Table 2. A summary of available heterogeneity-robust DID estimators.

Panel A: Estimators ruling out dynamic effects			
<i>(Can be used when outcome unaffected by past treatments)</i>			
Treatment	Estimators available	Stata commands	See:
Binary	de Chaisemartin and D'Haultfœuille (2020)	did_multiplegt	3.1
	Imai and Kim (2021)		3.1
	Borusyak et al. (2021)	did_imputation	3.2.3
Discrete	de Chaisemartin and D'Haultfœuille (2020)	did_multiplegt	3.1
Continuous, with stayers	de Chaisemartin et al. (2022)	See Section 3.1	3.1
Continuous, without stayers	de Chaisemartin et al. (2022)	See Section 3.1	3.1
	Graham and Powell (2012)	gmm	3.1
	Chamberlain (1992)	gmm	3.1
Several treatments	de Chaisemartin and D'Haultfœuille (2021b)	did_multiplegt	3.3
Panel B: Estimators allowing dynamic effects			
<i>(Can be used when outcome affected by past treatments)</i>			
Treatment	Estimators available	Stata commands	See:
Binary and staggered	Callaway and Sant'Anna (2021)	csdid	3.2.1
	Sun and Abraham (2021)	eventstudyinteract	3.2.2
	Borusyak et al. (2021)	did_imputation	3.2.3
Binary or discrete, nonstaggered	de Chaisemartin and D'Haultfœuille (2021a)	did_multiplegt	3.3
	de Chaisemartin and D'Haultfœuille (2021a)	did_multiplegt	3.3
Continuous and staggered	de Chaisemartin and D'Haultfœuille (2021a)	did_multiplegt	3.3
	Callaway et al. (2021)		3.3
Continuous and nonstaggered, with stayers	de Chaisemartin et al. (2022)	See paper	3.3
Continuous and nonstaggered, without stayers	No estimator available yet		
Several treatments	de Chaisemartin and D'Haultfœuille (2021b)	did_multiplegt	3.3

Notes: All the Stata commands have R equivalents with the same name, except `eventstudyinteract` that does not have an R equivalent, and `csdid` whose R equivalent is called `did`. The table's last column indicates the section of the paper where the estimator is described.

Questions? Comments?

Thank you!

References

Heavily based on Claire Palandri's 2022 version and Anna Papp's 2024 version of the Causal Inference Workshop.

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